

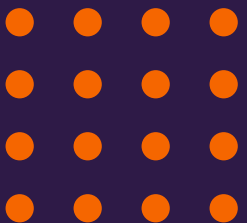


Perceptron and Multilayer Perceptron

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College of
**ENGINEERING, COMPUTING
AND APPLIED SCIENCES**





Lecture Outline

- Introduction to Artificial Neural Networks (ANNs)
- Review of Classification and Logistic Regression
- Single Neuron Network ('Perceptron')
- Multilayer Perceptron (MLP)

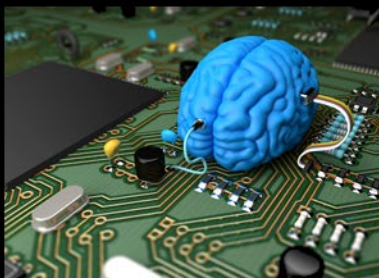


Artificial Neural Networks

Deep Learning



What society thinks I do



What my friends think I do



What other computer
scientists think I do



What mathematicians think I do



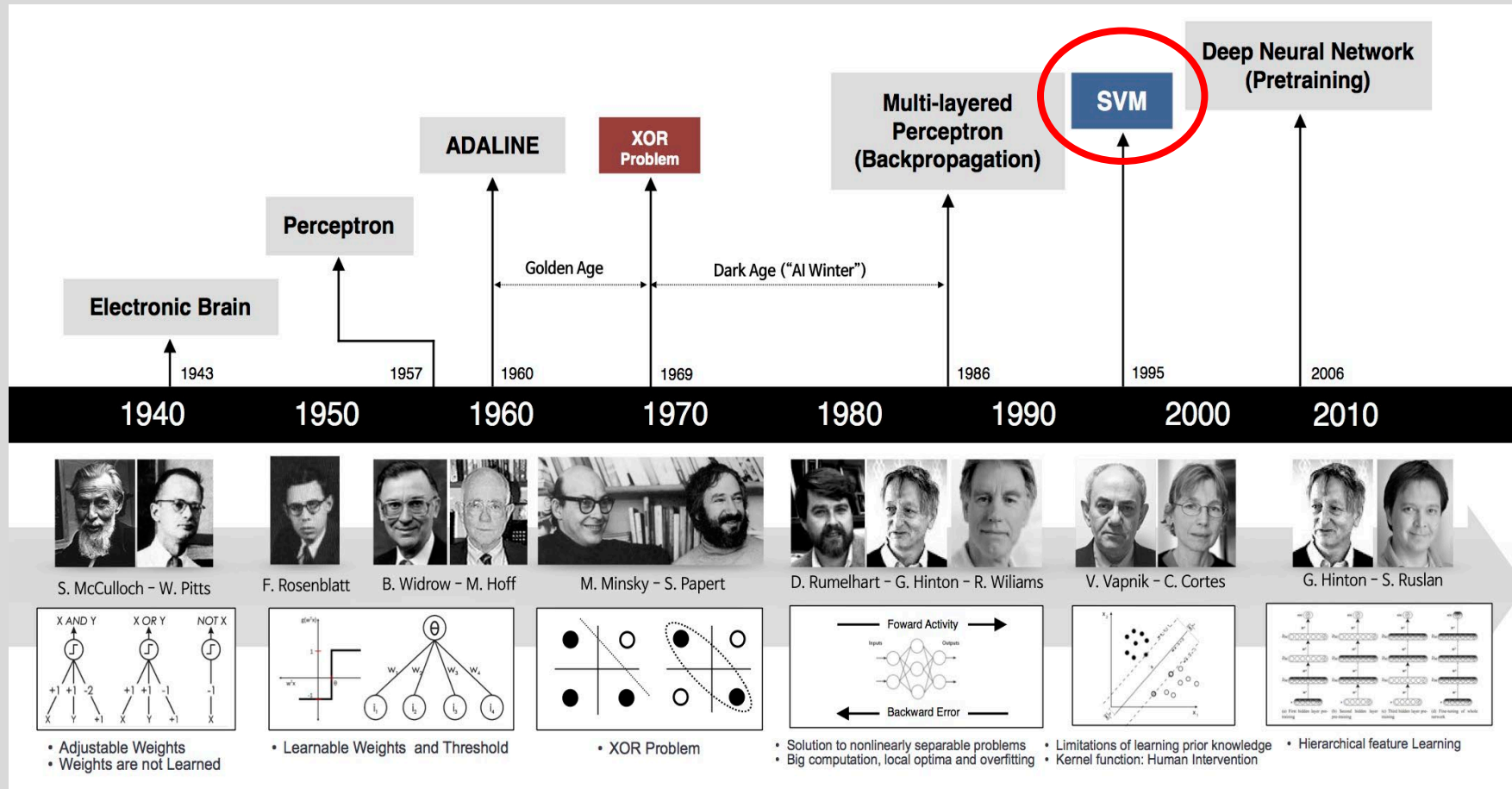
What I think I do

```
In [1]:  
import keras  
Using TensorFlow backend.
```

What I actually do

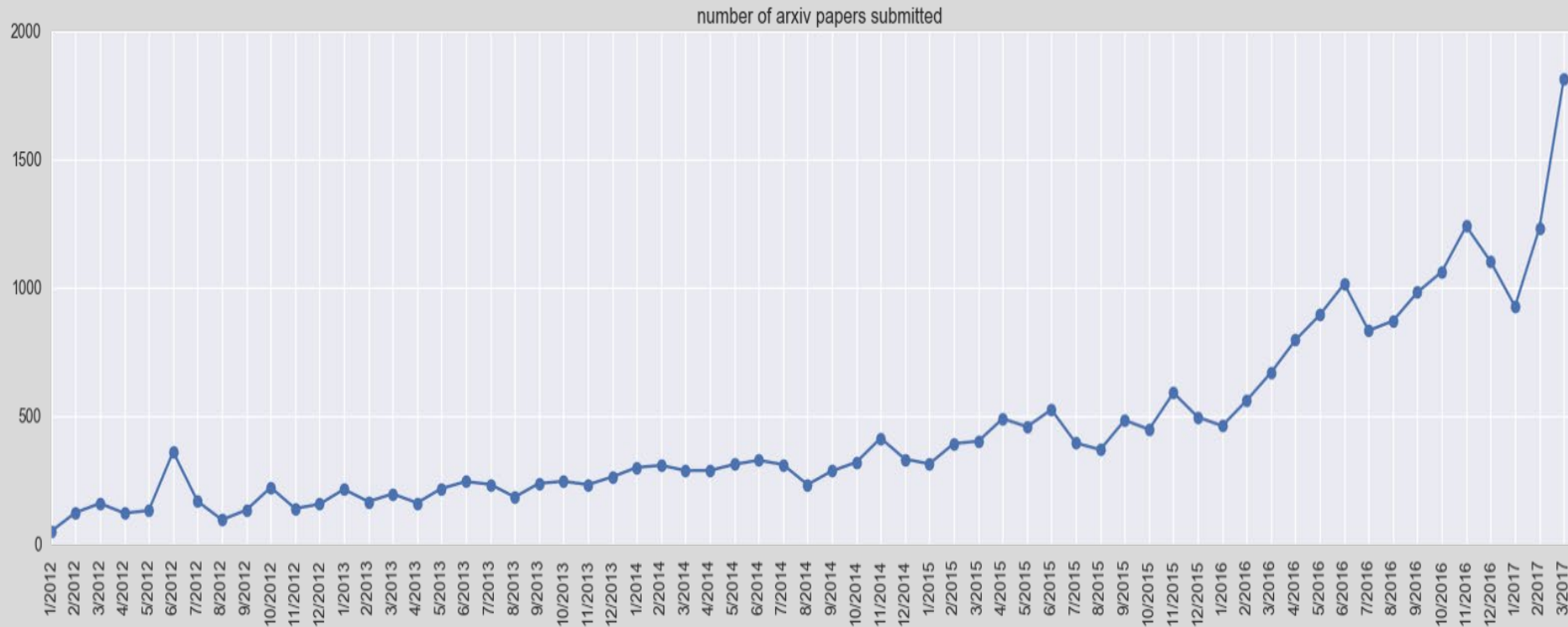


Historical Trends





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Regression and Classification

Methods that are centered around modeling and prediction of a *quantitative* response variable (ex, number of taxi pickups, number of bike rentals, etc.) are called **regressions** (and Ridge, LASSO, etc.).

When the response variable is *categorical*, then the problem is no longer called a regression problem but is instead labeled as a **classification problem**.

The goal is to attempt to classify each observation into a category (aka, class or cluster) defined by Y , based on a set of predictor variables X .



Typical Classification Examples

The motivating examples for this lecture(s), homework, and labs are based on classification. Classification problems are common in these domains:

- Trying to determine where to set the *cut-off* for some diagnostic test (pregnancy tests, prostate or breast cancer screening tests, etc...)
- Trying to determine if cancer has gone into remission based on treatment and various other indicators
- Trying to classify patients into types or classes of disease based on various genomic markers



Heart Data

response variable Y
is Yes/No

Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR	ExAng	Oldpeak	Slope	Ca	Thal	AHD
63	1	typical	145	233	1	2	150	0	2.3	3	0.0	fixed	No
67	1	asymptomatic	160	286	0	2	108	1	1.5	2	3.0	normal	Yes
67	1	asymptomatic	120	229	0	2	129	1	2.6	2	2.0	reversable	Yes
37	1	nonanginal	130	250	0	0	187	0	3.5	3	0.0	normal	No
41	0	nontypical	130	204	0	2	172	0	1.4	1	0.0	normal	No

Heart Data

These data contain a binary outcome HD for 303 patients who presented with chest pain. An outcome value of:

- **Yes:** indicates the presence of heart disease based on an angiographic test,
- **No:** means no heart disease.

There are 13 predictors including:

- Age and Sex
- Chol (a cholesterol measurement),
- MaxHR and RestBP

and other heart and lung function measurements.



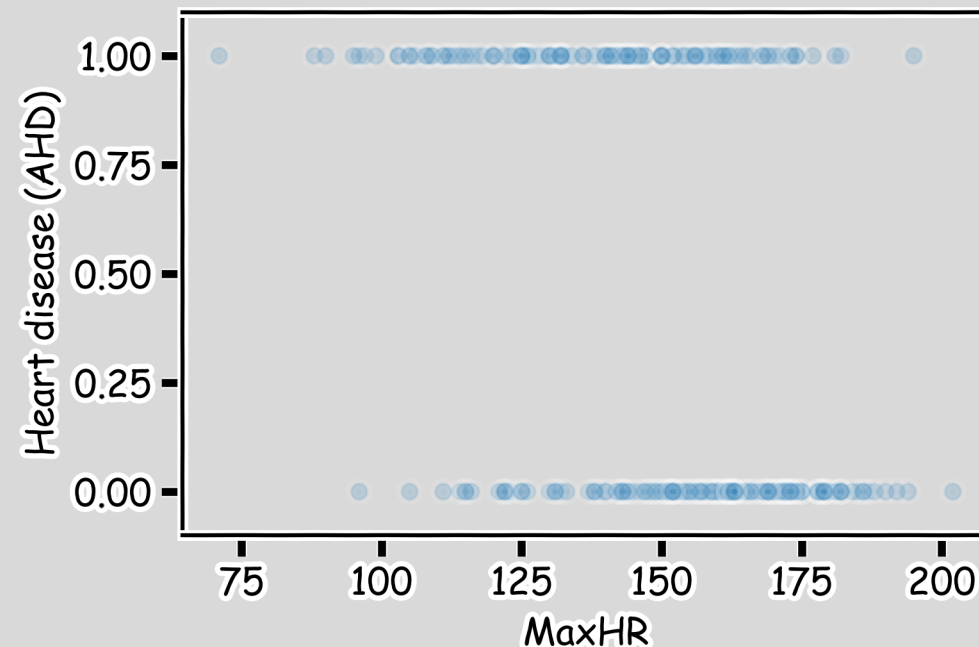
Logistic Regression

Logistic Regression addresses the problem of estimating a probability, $P(y = 1)$, given an input X . The logistic regression model uses a function, called the *logistic* function, to model $P(y = 1)$:

$$P(Y = 1) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$

Heart Data: Logistic Estimation

We'd like to predict whether or not a person has a heart disease. And we'd like to make this prediction, for now, just based on the MaxHR.



Logistic Regression

As a result, the model will predict $P(y = 1)$ with an S -shaped curve, which is the general shape of the logistic function.

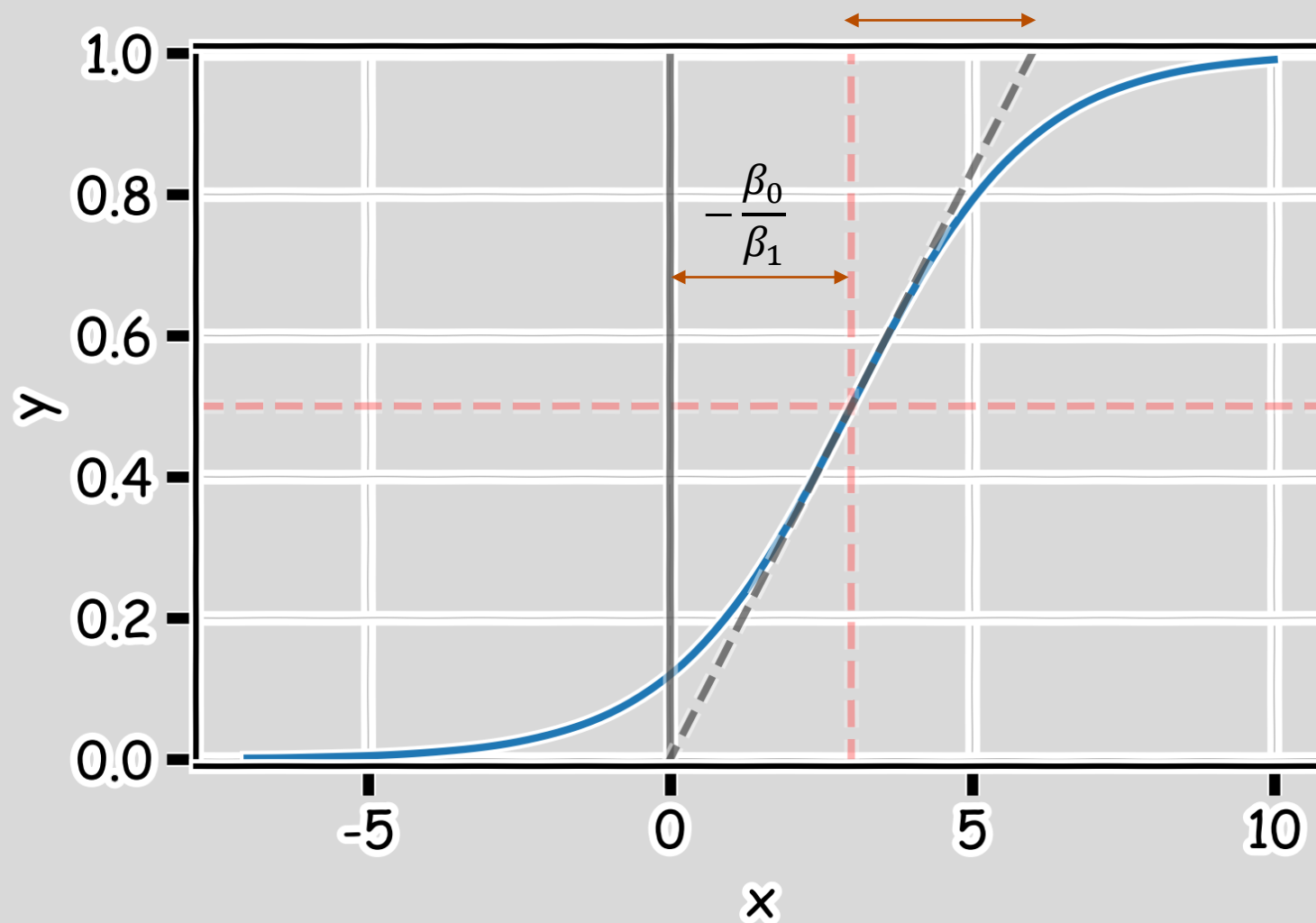
β_0 shifts the curve right or left by $c = -\frac{\beta_0}{\beta_1}$.

β_1 controls how steep the S -shaped curve is distance from $\frac{1}{2}$ to ~ 1 or $\frac{1}{2}$ to ~ 0 to $\frac{1}{2}$ is $\frac{2}{\beta_1}$

Note: if β_1 is positive, then the predicted $P(y = 1)$ goes from zero for small values of X to one for large values of X and if β_1 is negative, then has the $P(y = 1)$ opposite association.



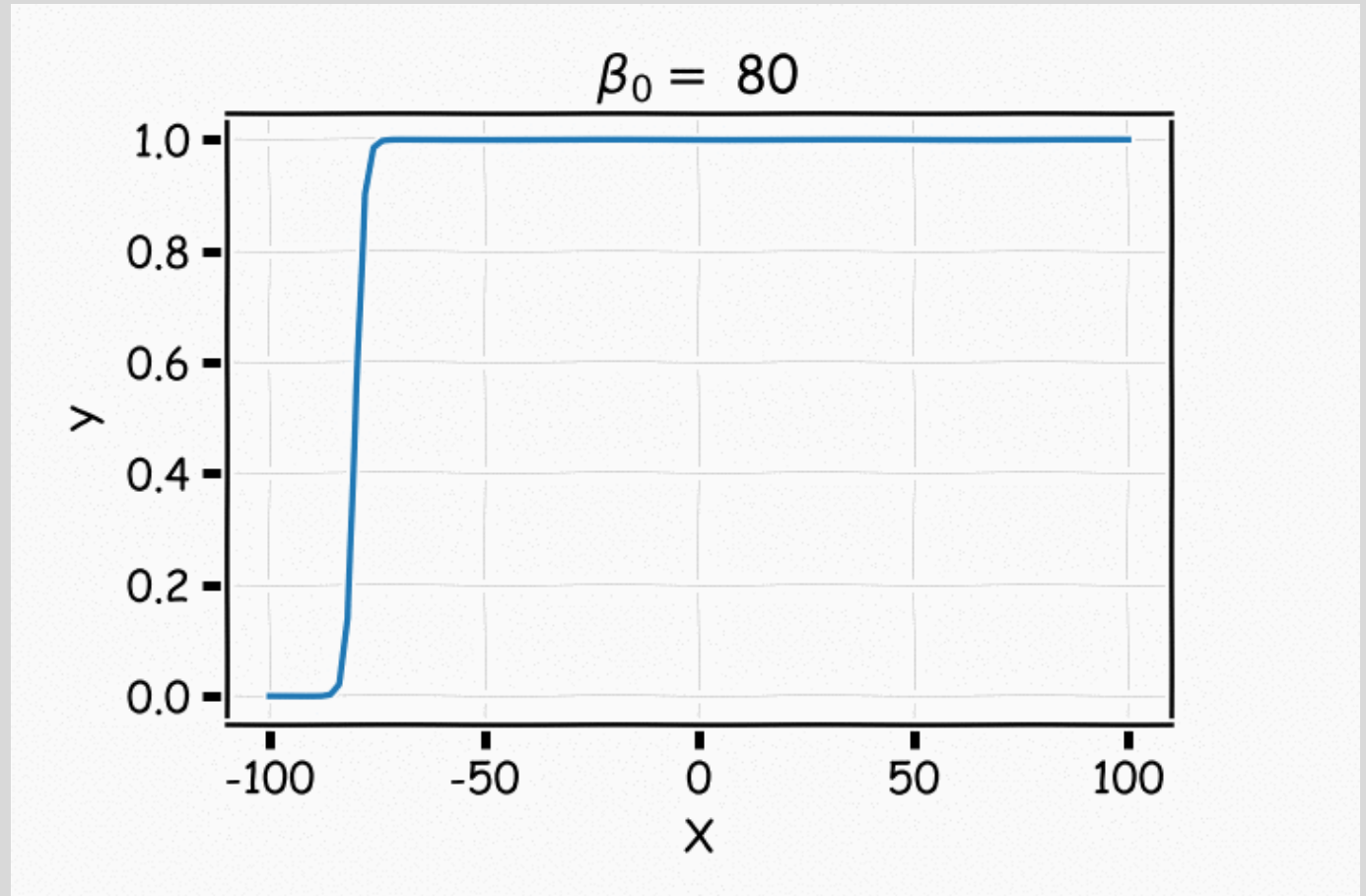
Logistic Regression





Logistic Regression

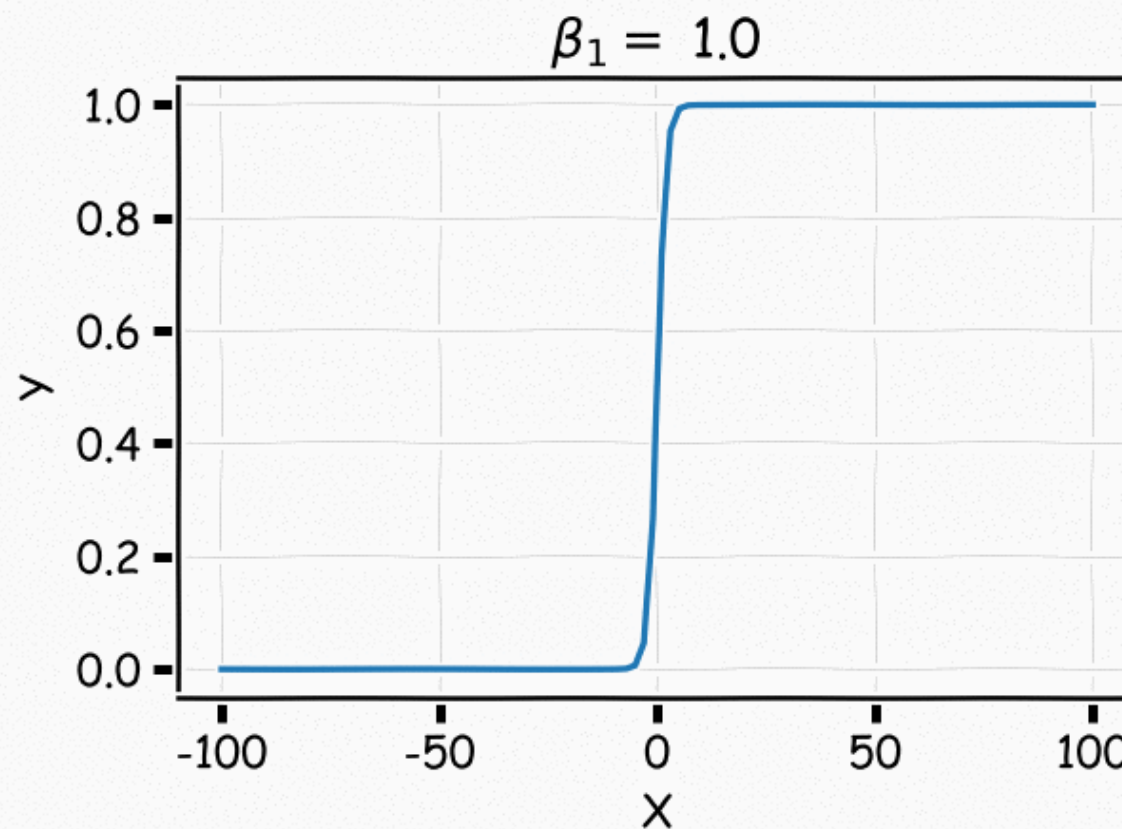
$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$





Logistic Regression

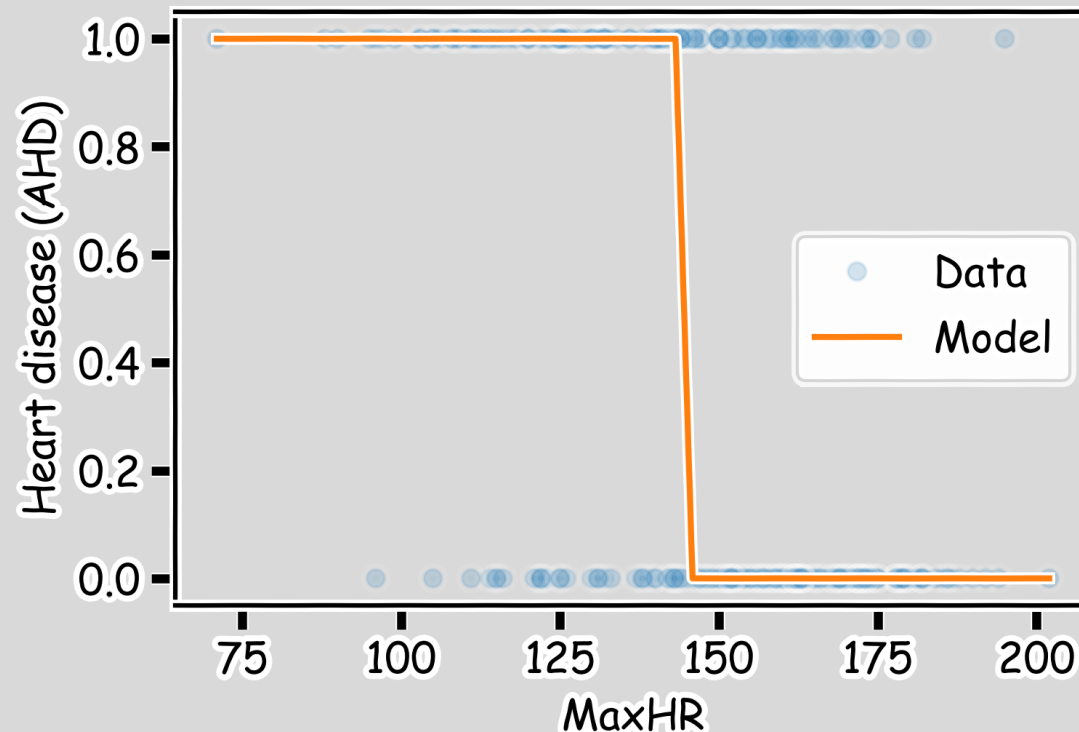
$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}}$$



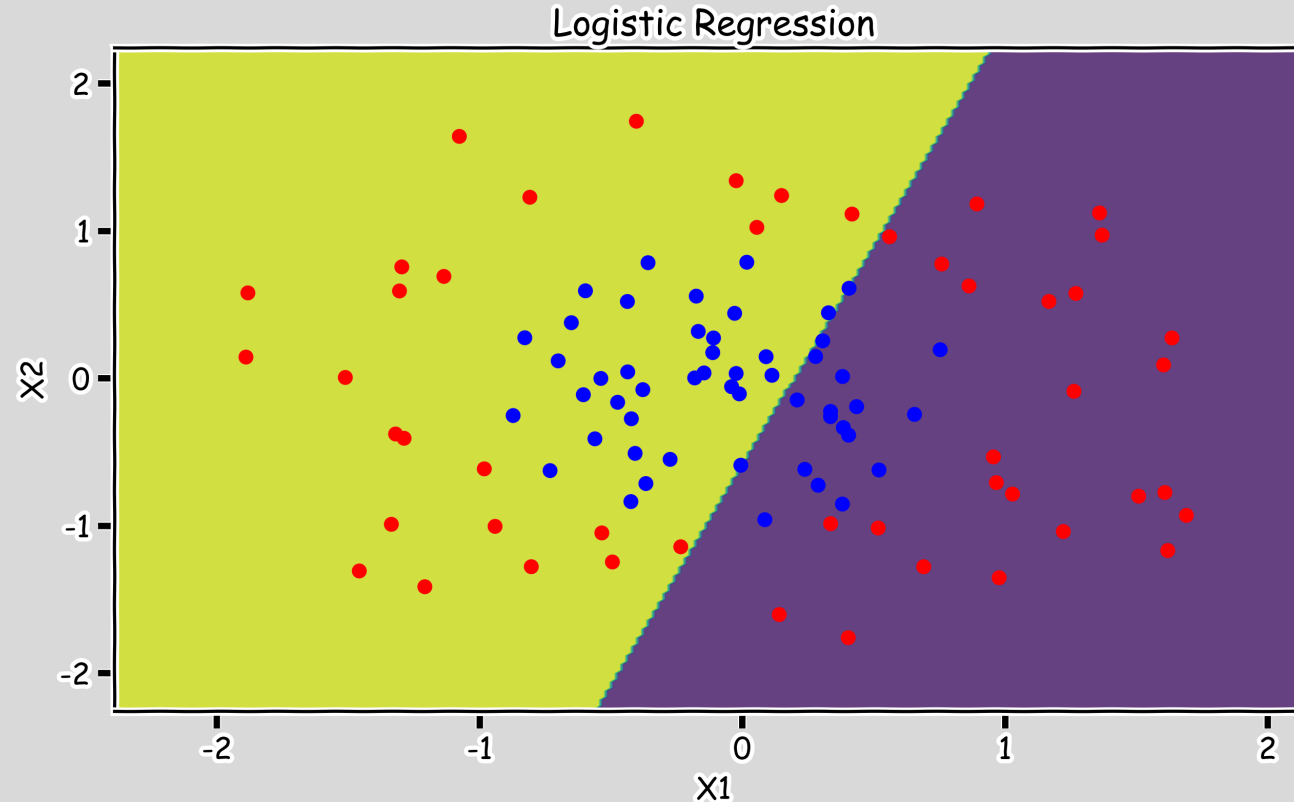
Logistic Regression Coefficient Estimation

Find the coefficients that minimize the loss function

$$\mathcal{L}(\beta_0, \beta_1) = - \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$

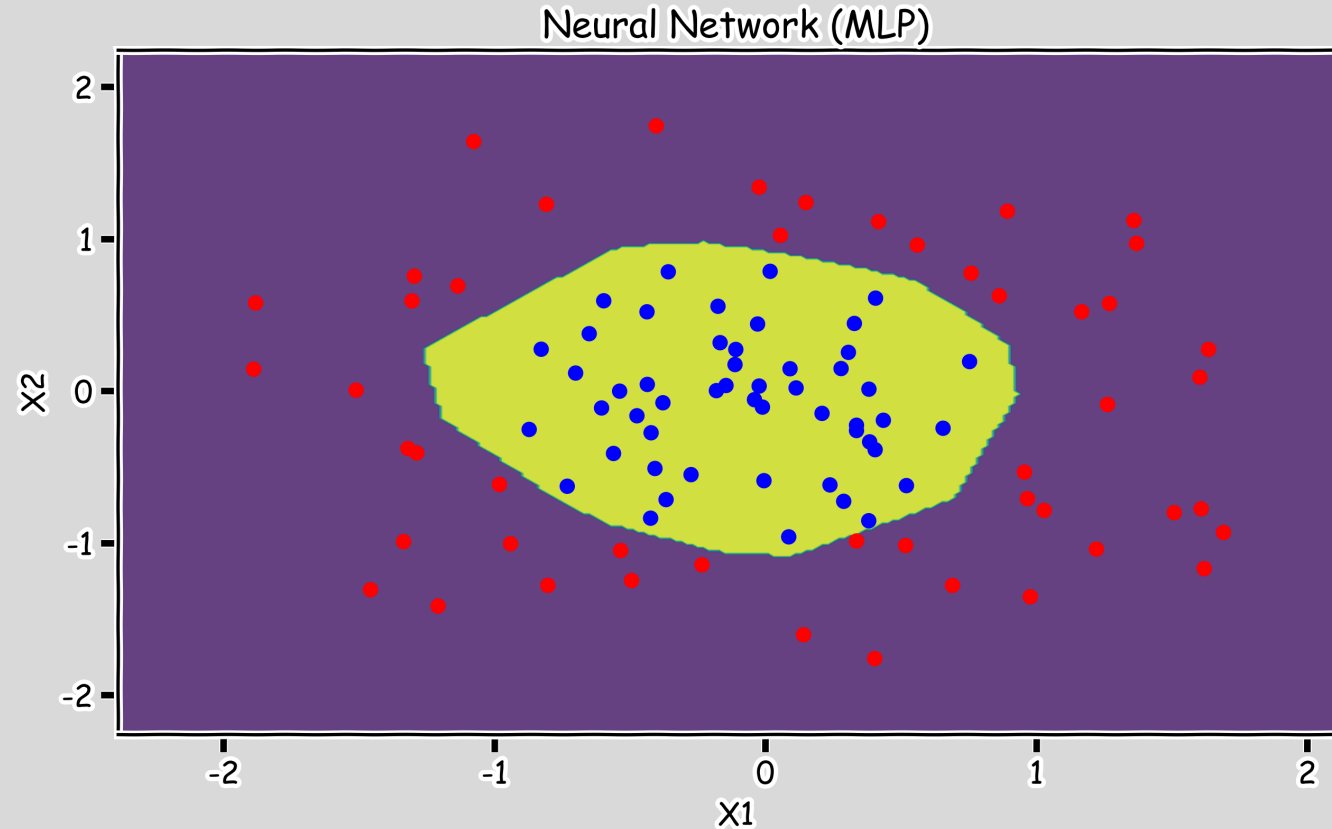


Need for Non-Linearity



Without augmenting the features (i.e. without adding X_1^2 or X_2^2 non-linear features), Logistic Regression is incapable of modeling the correct decision boundary.

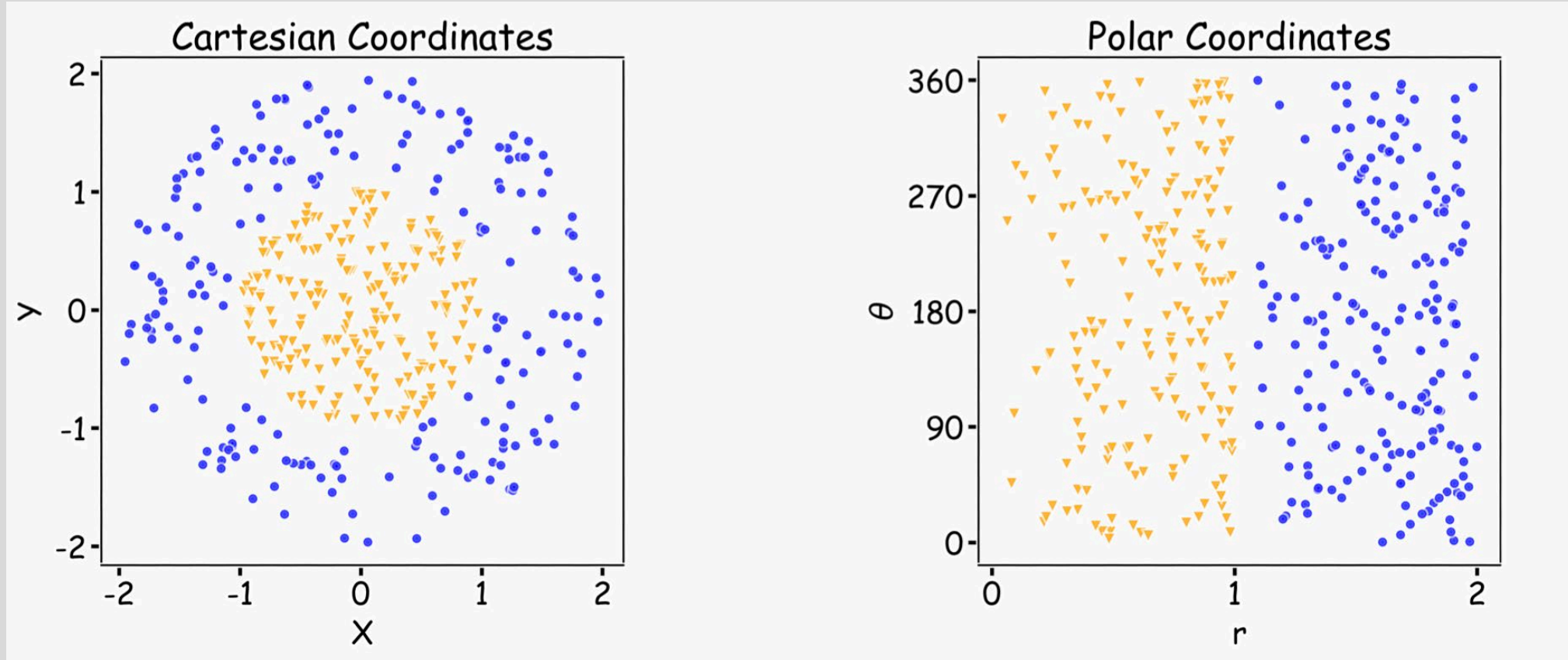
Need for Non-Linearity



A neural network is a powerful non-linear model that can easily model the non-linear decision boundary correctly.



Representation Matters



Neural networks can **learn useful representations** for the problem. This is another reason why they can be so powerful!